Midterm 2

Graded

Student

Marcus Huang

Total Points

109 / 150 pts

Question 1

T/F/U — **36.5** / 60 pts

1.1 a — **5.5** / 10 pts

- **– 0 pts** Correct
- ✔ **– 1.5 pts** Some good explanation that decreasing the corporate tax rate can either increase investment or not have an effect (with full immediate deductions) depending on the model
- **– 3 pts** Misses that decreasing the corporate tax rate can either increase investment by reducing the tax burden on this compared to other ways of earning a return or not have an effect as the k that maximizes profit is the same one that maximizes positive number * profit (with full immediate deductions) depending on the model
- **– 1.5 pts** Some good explanation that investment being immediately deductible raises the present value of the write-offs, increasing investment
- ✔ **– 3 pts** Misses that investment being immediately deductible raises the present value of the write-offs, increasing investment
- **– 10 pts** Misses that decreasing the corporate tax rate can either increase investment or not have an effect (with full immediate deductions) depending on the model and misses that investment being immediately deductible raises the present value of the write-offs, increasing investment (**often adds little beyond saying true and repeating the statement**)

1.2 b — **0** / 10 pts

- **– 0 pts** Gives a full, correct explanation (false, as this is a pecuniary externality/through normal workings of price mechanism/is a shift out in demand, so does not represent an externality or market failure)
- **– 3 pts** Answer gives a good and correct explanation but does not fully explain (along lines of false, as this is a pecuniary externality/through normal workings of price mechanism/is a shift out in demand, so does not represent an externality or market failure)
- **– 6 pts** Typically perceives this as a standard negative externality in production and must state "this is a harm to others not involved in the transaction" with a well-developed explanation, but fails to recognize that this in fact represents a shift in demand
- ✔ **– 10 pts** Incorrect/insufficient for credit (often refers to the situation as an externality with terms used imprecisely)

1.3 c — **9** / 10 pts

- **– 0 pts** Correct answer with complete explanation
- ✔ **– 1 pt** Correct answer with solid explanation
- **– 3 pts** Solid answer, but some components of the weitzman-style analysis are missing or incorrect
- **– 5 pts** Uses Weitzman framework but has significant flaws or gaps in logic (e.g., argues for opposite answer)
- **– 5 pts** Coherently discusses flat MD and/or uncertain costs but doesn't use weitzman framework
- **– 6 pts** Argues that because we don't know price, we should set a quantity limit. Incorrect because without knowing the MC we also can't know the optimal quantity
- **– 7 pts** Large omissions or errors in response (e.g., doesn't discuss flat MD and uncertain costs in response, confuses MD and total damages, draws on frameworks that don't answer the question)
- **– 10 pts** Incomplete or insufficient answer for credit

1.4 d — **8** / 10 pts

- **– 0 pts** Correct answer with complete explanation
- ✔ **– 2 pts** Correct answer (noting that we sum for public goods but for private we just use individual MRS) but without explaining why there is a difference in how we treat public v private goods
- **– 3 pts** Correct answer: accurately explains why they are different without explaining how the calculation differs (how do we find social efficiency for private goods if we don't sum?)
- **– 4 pts** Correct answer but incomplete explanation: (e.g., discusses key differences between public/private but insufficient discussion of what this means for social efficiency; only explains the rule for one kind of good)
- **– 5 pts** Correct conclusion but incorrect explanation: inaccurate description of a public vs private good
- **– 6 pts** Correct conclusion but incorrect explanation: focuses on irrelevant or incorrect features of the market (e.g., "substitutability" of goods)
- **– 7 pts** Correct conclusion but very vague explanation (e.g., doesn't focus on difference between private and public goods)
- **– 8 pts** Incorrect answer AND explanation
- **– 9 pts** Says false only
- **– 10 pts** Incomplete or insufficient for credit

1.5 e — **10** / 10 pts

- ✔ **– 0 pts** Correct
- **– 6 pts** does not demonstrate understanding that giving itself provides utility (warm-glow) or restates or confuses warm glow with altruism
- **– 3 pts** vaguely mentions that people derive utility from giving itself
- **– 2 pts** missing tie in to crowd-out is less than one-for-one
- **– 10 pts** blank or insufficient info for credit

1.6 f — **4** / 10 pts

- **– 0 pts** Correct
- ✔ **– 6 pts** missing or incorrect understanding of imperfect information or restate
- **– 3 pts** suggests uncertainty/imperfect info but vague
- **– 2 pts** does not connect to Tiebout model (can also be earned by correctly using empirical evidence that demonstrates imperfect information)
- **– 10 pts** blank or insufficient info for credit

Question 2

Empirical Application

24.5 / 30 pts

2.1 **a** — **6.25** / 7.5 pts

- **– 0 pts** Fully explains that the idea is a diff-in-diff based on exposure to welfare reform in terms of % of non-citizens in county and that an upward sloping relationship is consistent with the idea that counties more exposed to welfare cuts experienced larger increases in church charitable spending
- ✔ **– 1.25 pts** Partially explains that the idea is a diff-in-diff based on exposure to welfare reform in terms of % of non-citizens in county
- **– 2.5 pts** Does not explain that the idea is a diff-in-diff based on exposure to welfare reform in terms of % of non-citizens in county (often writes y variable positively correlated with x variable without an explicit link to DiD)
- **– 1.25 pts** Partially explains that an upward sloping relationship is consistent with the idea that counties more exposed to welfare cuts experienced larger increases in church charitable spending
- **– 2.5 pts** Does not explain that an upward sloping relationship is consistent with the idea that counties more exposed to welfare cuts experienced larger increases in church charitable spending
- **– 7.5 pts** Insufficient for credit (e.g. often writes that y variable positively correlated with x variable without linking explicitly to diff in diff)

2.2 **b** — **4.5** / 7.5 pts

Do you find the causal interpretation convincing?

- **– 0 pts** Not convincing; points to confounding variables or or different trends
- ✔ **– 1 pt** Not convinced; argues that it isn't causal without discussing confounders or parallel trends
- **– 2.5 pts** Not convinced; no explanation or incorrect reasoning
- **– 2.5 pts** Is convinced; refers to some possibility of alternative mechanisms
- **– 2.5 pts** is convinced; assumes parallel trends when this isn't a given
- **– 3 pts** Is convinced; says the trend is consistent with the theorized effect without considering potential confounders or different trends
- **– 3.5 pts** Is convinced; no explanation
- **– 4 pts** Doesn't discuss whether they are convinced

Additional evidence

- **– 0 pts** Wants a graph with pre-trends or other evidence to support DiD assumptions
- **– 2 pts** Wants evidence on potentially relevant factors but does not tie it directly to the DiD assumptions
- ✔ **– 2 pts** Suggests using a DiD approach but doesn't speak about evidence they want to support that approach
- **– 3 pts** Suggests evidence that is too vague, would undermine the conclusion, or otherwise demonstrates a lack of understanding of the problem
- **– 3.5 pts** Doesn't discuss any specific additional evidence they want to see

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- **– 7.5 pts** Blank

2.3 **c** — **7.5** / 7.5 pts

- ✔ **– 0 pts** all correct
- **– 1 pt** all correct but added error term
- **– 2 pts** one wrong
- **– 4 pts** two wrong
- **– 6 pts** three wrong
- **– 7.5 pts** four wrong

2.4 **d** — **6.25** / 7.5 pts

- **– 0 pts** 1) The difference-in-differences estimate equals the coefficient δ, AND
  2) interpretation with context: church charitable spending increased **more** in treatment counties (those more exposed to the welfare cuts) **than** in control counties **after** the reform.
- **– 3.25 pts** wrong coefficient
- **– 3.25 pts** interpretation is wrong, missing, insufficient, just restates positive correlation
- ✔ **– 1.25 pts** interpretation mostly correct: typically does not relate it to the control group.
- **– 6.5 pts** wrong on both fronts but attempted
- **– 7.5 pts** no attempt

Question 3

Analytic **48** / 60 pts

3.1 a **9** / 12 pts

− 0 pts needs all below for full credit:

- phase out when z = 40 (2 points)
- segment 1: c(z) = 20 + 0.5z (2 points)
- segment 2: c(z) = z (2 points)
- a perfectly **drawn** budget constraint with: (4 points)
  - correct y-intercept (0, 20)
  - slope at phase out of 0.5
  - correct kink point (40, 40)
  - slope after kink point of 1
- two free points for attempt

− 2 pts incorrect phase out when z = 40 (2 points)

− 2 pts incorrect segment 1: c(z) = 20 + 0.5z

− 2 pts incorrect segment 2: c(z) = z

− 4 pts four mistakes on budget constraint

✔ **− 3 pts** three mistakes on budget constraint

− 2 pts two mistakes on budget constraint

− 1 pt one mistake on budget constraint

− 1 pt this is for a select few cases:

- did not draw the budget constraint but write down all four components needed
- do not provide after-tax-and-transfer income for segment (s) but budget constraint is entirely correct

− 12 pts blank

3.2 b **10** / 12 pts

− 0 pts Full credit :

- 50% participation tax rate for worker @ z = 20
- 25% participation tax rate for worker @ z = 80
- low earner faces the higher participation tax rate
- provides correct intuition: the benefits lost ("cost") from entering the labor force represent a larger share of their income than high earners

(everyone gets 2 points for an attempt to account for math errors that lead to wrong answers)

− 2.5 pts not 50% participation tax rate for worker @ z = 20

− 2.5 pts not 25% participation tax rate for worker @ z = 80

− 1 pt does not say low earner faces the higher participation tax rate

− 4 pts missing or off-topic intuition (focuses on the mechanics)

✔ **− 2 pts** wrong intuition, typically:

- wrong direction (i.e, encourages work)
- discusses income effect (how much to work) rather than to work or not

− 1 pt intuition is incomplete

− 10 pts incorrect but an attempt

− 12 pts incorrect and no attempt

3.3 c **12** / 12 pts

Final answer

✔ **+ 3 pts** Correct final answer: $9000

+ 0 pts Final answer missing or incorrect

Work

✔ **+ 9 pts** Shows work thoroughly, correct

+ 7 pts Shows work thoroughly but incorrect

+ 4 pts Some work shown but incorrect

+ 0 pts No work shown

+ 2 pts Flex points

3.4 d **9** / 12 pts

Drawing

✔ **+ 3 pts** Drew graph with correct structure

+ 1.5 pts Drew graph with key elements missing or incorrect

+ 0 pts Does not draw graph

Coordinates

✔ **+ 3 pts** Correctly labels coordinates

+ 1.5 pts Incorrect or incomplete coordinates

+ 0 pts Did not label coordinates

Slopes

+ 3 pts Correct slopes

+ 1.5 pts Incorrect or incomplete slopes

✔ **+ 0 pts** Missing slopes

Phase-out level

✔ **+ 3 pts** Does identify phase out = 80

+ 1.5 pts Incorrect phase out level

+ 1.5 pts Gets the phaseout right but doesn't explicitly point it out

+ 0 pts Does not identify earnings level that phases out credit

+ 0 pts Blank

+ 2 pts Flex points

3.5 **e.i** **4** / 4 pts

- ✔ **− 0 pts** Correct - 25% and half of the previous participation tax rate
- **− 2 pts** One out of 25% and halves incorrect
- **− 3 pts** Typically says participation tax rate is lower (not half) but gets 25% incorrect
- **− 4 pts** Incorrect

3.6 **e.ii** **4** / 4 pts

- ✔ **− 0 pts** Correct
- **− 2 pts** One of extra cost for one worker =$5 or total mechanical cost = $500 incorrect
- **− 4 pts** Both of extra cost for one worker =$5 and total mechanical cost = $500 incorrect

3.7 **e.iii** **0** / 4 pts

- **− 0 pts** Correct
- **− 1 pt** Gets free lunch intuition incorrect or gets extensive margin saving incorrect or gets percent increase in net of tax rate incorrect but gets at least 1 net of tax rate correct
- **− 2 pts** Correctly identifies 50% increase in net of tax rate but gets extensive margin saving and free lunch intuition incorrect (or vice versa)
- **− 3 pts** Typically finds at least one net of tax rate correctly but not the percentage change
- ✔ **− 4 pts** Incorrect

Econ 131
Spring 2026
Danny Yagan

Midterm 2

Monday, April 6, 2026

Exam Instructions:

- The exam lasts 75 minutes.
- This exam has three questions. Question 1 and Question 3 are worth 60 points each, and Question 2 is worth 30 points.
- Total points for the exam is 150 points.
- Explanation should be written using pens (we recommend black or blue ink, as these scan the best).
- Write your solutions in the spaces provided.
- If you need more room to write your answers or need to re-draw a graph, use the extra pages at the end. Make sure to note clearly if your solutions continue on a different page.
- Do not write your solutions on pages that say "Do not write on this page". Answers written on these pages will not be graded.
- Show your work. Credit will only be awarded on the basis of what is written on the exam.
- Sign the academic honesty pledge. If you do not affirm this pledge, your exam will be marked <u>invalid</u>.

Student Name: Ziyu Huang (Marcus Huang)

Student ID Number: 3041432436

Affirm the academic honesty pledge below. For those writing on a non-printed copy, please just write "Academic Honesty Pledge as on exam", and sign your name.

If you do not affirm this pledge, your exam will be marked invalid.

0. ACADEMIC HONESTY PLEDGE

I confirm that I have abided by all academic honesty rules for UC Berkeley and Economics 131. I confirm that I did not see this exam before my official exam start time. I confirm that I have not shared and will not share this exam with anyone else. I confirm that I haven't copied from anybody else's exam.

Signature:

1. True/False/Uncertain Questions (60 points total)

You can start your answer "True", "False", "Uncertain", "Partially true", or "Partially false", etc., but your choice of the label does not matter much. What matters is your explanation. **Note that it is possible to answer each question for full credit with three sentences or fewer, and answers longer than ten lines long will not be graded.**

(a) Suppose that a tax reform cuts the corporate tax rate and makes investment immediately deductible, rather than deductible over time. The tax reform will increase investment.

True, tax reform make the corporate tax rate lower and deduct investment, people will invest more for more benifits.

(b) Suppose a town's new data center uses a lot of electricity, which makes electricity more expensive for the town's families. This is an example of a market failure.

True, this is a negative externality. The data center uses lots of electricity that make differences on town's families spense on it. Families have to pay more.

(c) When pollution abatement costs are uncertain and when the marginal damage from pollution does not depend on the pollution level, tradable permits can be preferable to a corrective tax.

MD from pollution does not depend on the pollution level

⇓

almost fixed

⇓

MD is flat

⇓

tax is better than permit in this case

↓

for steep MD.

(d) For private goods, social efficiency requires summing marginal rates of substitution across consumers in the same way as for public goods. False.

public goods : sum of all individuals' MRS.

private goods : focus on each individual's MRS

Charity.

(e) Warm-glow motives can imply less than one-for-one crowd-out of private giving by government provision.

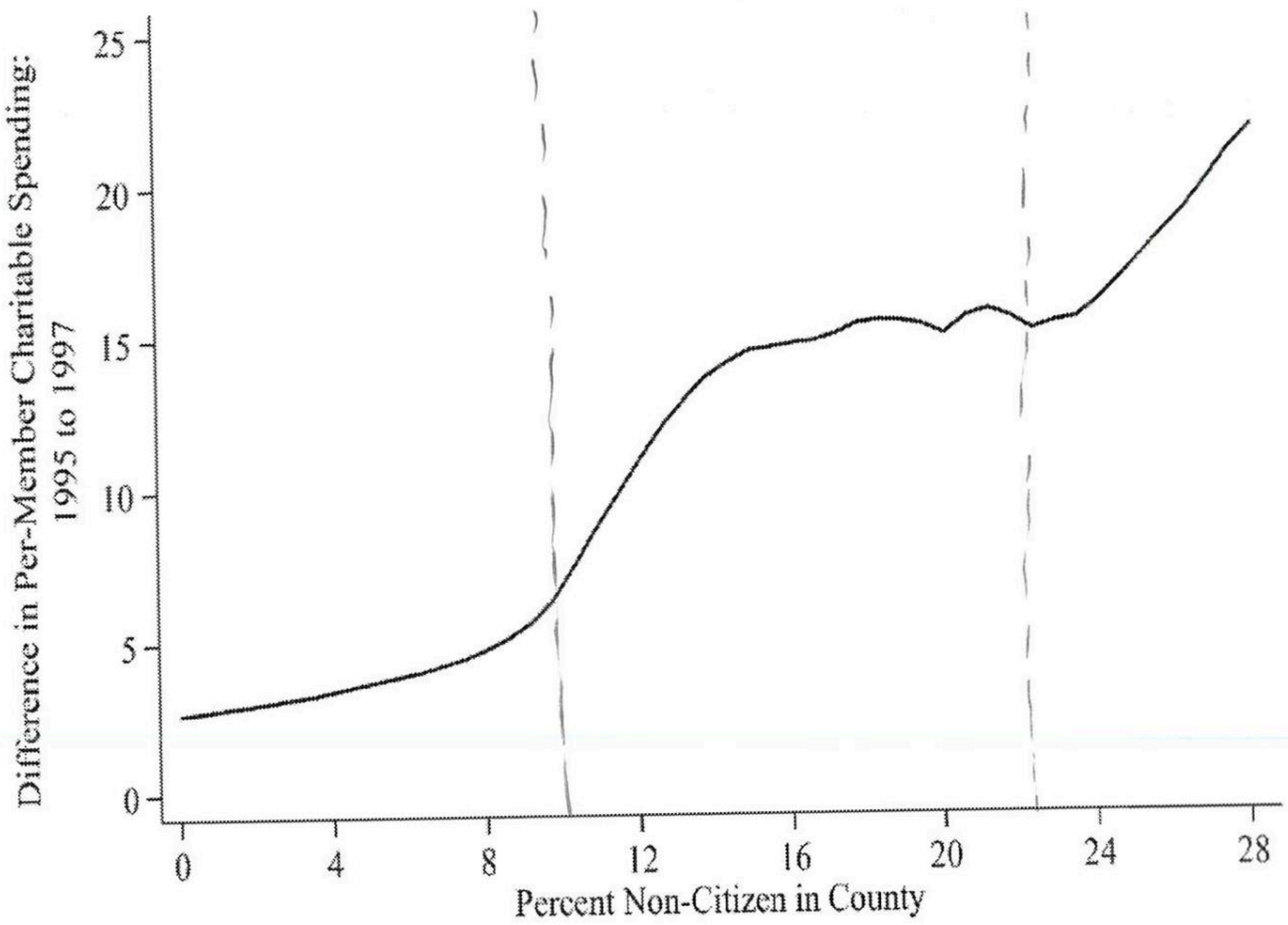
No matter how the government set the taxes by provisions, in real world, empirically, there still somebody doing donation or private giving. Warm-glow is more behavioral and one-for-one crowd-out is more mechanical.

(f) People know the quality of local public goods.

Sort of. people vote by their feet.

2. Empirical Application: Hungerman (2005) and Crowd-Out of Church Charity (30 points total)

In 1996, welfare reform reduced or eliminated some transfers for non-citizens. Hungerman (2005) studies whether church-provided charitable spending rose more in places more exposed to this reform. The figure below plots the change in per-member charitable spending from 1995 to 1997 against the percent non-citizen in the county.



Source: Hungerman 2005

(a) Explain in words the difference-in-differences style conclusion this figure is trying to suggest. What broad pattern in the figure would be consistent with the idea that counties more exposed to the welfare cuts experienced larger increases in church charitable spending? *(7.5 points)*

— when doing Difference-in-Differences analytic, we care about the trends more than specific levels.

—Based on the figure, the per-member charitable spending was going up while the percent non-citizen in the conty increases in genero

— The welfare reform basically cut the transfer for the non-citizen people, which make them poorer. The higher pecent non-citizen a county has, the more exposed to this reform. Then the Church want to make them live a better life or back to the living quality as before, they spend more money.

— In conclusion, according to the trend shown on the figure, it can explain the question / issue we focus on and proof the study.

(b) Based on this figure alone, do you find the causal interpretation convincing? Answer yes or no and explain briefly. What additional evidence or alternative figure would you want to see to make the claim more convincing? *(7.5 points)*

No, the causal effect is related to the DD estimate.
which means we need to find the pre and post status.
the treatment group and the control group are needed
We need the curve for the control group which means it
had absent of the reform.

(c) Let $Treat_c = 1$ denote the counties with a high non-citizen share of the population, and let $Treat_c = 0$ denote otherwise. Let $Post_t = 1$ denote years after the reform and $Post_t = 0$ denote years before the reform. Let Y_{ct} denote church charitable spending in county c and year t. Suppose you estimate the regression

$$Y_{ct} = \alpha + \beta Treat_c + \gamma Post_t + \delta(Treat_c \times Post_t) + \varepsilon_{ct}.$$

Using the coefficients $\alpha, \beta, \gamma, \delta$ from this regression, fill in the four cells in the following table. Each cell equals a mean. For example, the control/pre cell equals the mean value of Y_{ct} in the control counties before the reform. *(7.5 points)*

	Pre ($Post_t = 0$)	Post ($Post_t = 1$)
Control ($Treat_c = 0$)	α	$\alpha + \gamma$
Treatment ($Treat_c = 1$)	$\alpha + \beta$	$\alpha + \beta + \gamma + \delta$

(d) Which regression coefficient or coefficients equal the difference-in-differences estimate? Briefly interpret what a positive value of that coefficient or coefficients would mean in this context. *(7.5 points)*

δ is the DD estimate.

This δ is positive will mean that Treatment does have a effect. we are counting the Church's spending on non-citizen, then this positive δ means this reform did make church do more charity.

3. Negative Income Tax vs. Earned Income Tax Credit (60 points total)

A city has 1,000 working-age residents. Each resident decides whether to work or not. A worker earns pre-tax labor income $z > 0$; a non-worker earns $z = 0$. Initially there are **no taxes or transfers.**

(a) The city introduces a **Negative Income Tax (NIT)**:

- Every resident receives a **guaranteed income of $20** (even with zero earnings).
- The guarantee is **phased out at 50 cents per dollar** of earnings.
- There are **no further taxes** above the point where the guarantee is fully phased out.

At what level of earnings z does the guarantee fully phase out? Write after-tax-and-transfer income $c(z) = z - T(z)$ for each segment, and draw the budget constraint with pre-tax earnings z on the x-axis and c on the y-axis. Label the y-intercept, any kink points (with coordinates), and the slope of each segment. *(12 points)*

50 cents phase out / dollar , total $20 → $20 / 50 cents

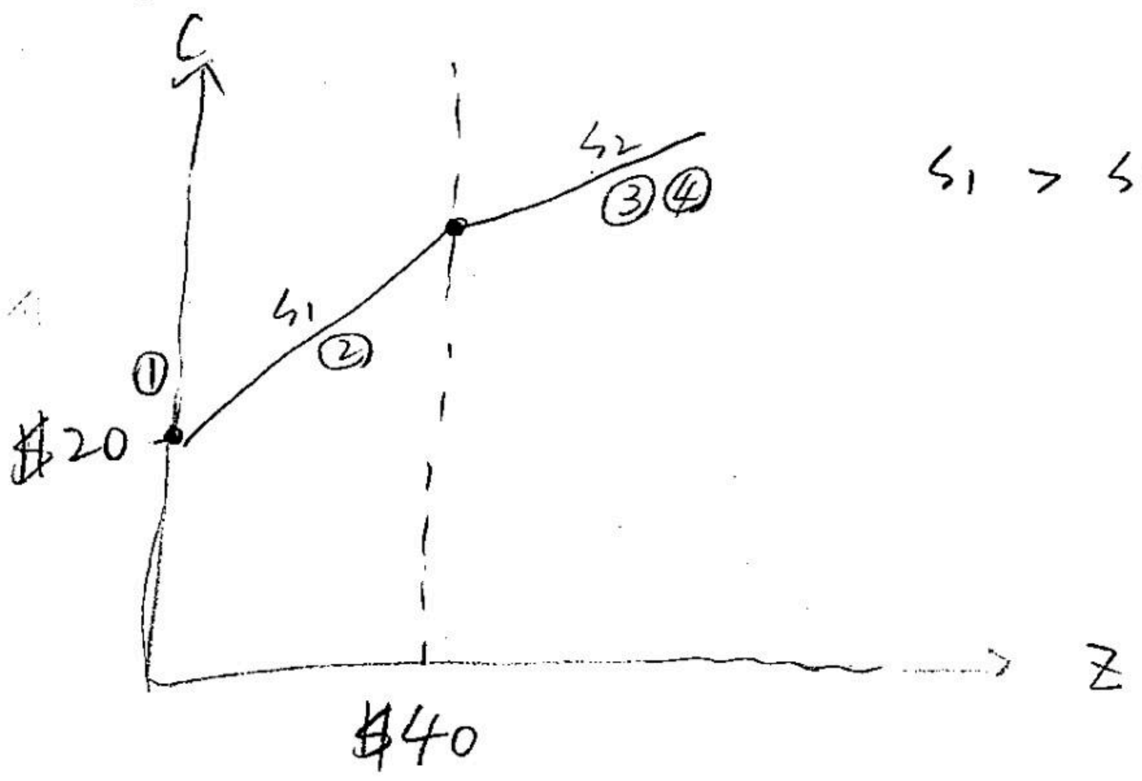
result : When earning $z = \$40$, the guarantee fully phase out.

① no income : $c_1(z) = \$20$

② income less than $40 : $c_2(z) = z - \frac{1}{2}z + \$20$

③ income equals $40 : $c_3(z) = z$

④ income larger than $40 : $c_4(z) = z$



$s_1 > s_2$ (slope).

(b) Using the NIT from part (a), compute the **participation tax rate**

$$\tau_p \ = \ \frac{T(z) - T(0)}{z}$$

for a worker earning $z = 20$ and for a worker earning $z = 80$. Which faces the higher participation tax rate? In one sentence, explain intuitively why a system with a phased-out guaranteed income creates especially high participation tax rates for low earners. *(12 points)*

① $T_p = \dfrac{T(20) - T(0)}{20} = \dfrac{10}{20} = 0.5$

② $T_p = \dfrac{T(80) - T(0)}{80} = \dfrac{20}{80} = 0.25$

a worker earning $z = 20$ faces a higher participation tax rate.

" They need money, low-income worker (under $40) get transfer.

⇒ they work more "

(c) Before the NIT was introduced, there were 700 workers and 300 non-workers. After the NIT is introduced:

① • 100 workers who had been earning $z = 20$ each **stop working**, discouraged by the high participation tax rate.

② • 500 workers who earn $z = 80$ each continue working.

③ • The remaining 100 workers who earn $z = 20$ each also continue working.

④ • The 300 original non-workers remain out of work.

Compute the government's **total cost** of the NIT (total grants paid out, net of any phase-out clawback). *(12 points)*

① government pay 100 X \$20 = \$2000 for that 100 workers

② pays 0

③ \$20 X 0.5 = \$10 , \$20 - \$10 = \$10 → 100 X \$10 = \$1000

④ 300 X \$20 = \$6000

thus : \$2000 + 0 + \$1000 + \$6000 = \$9000

total cost of the NIT

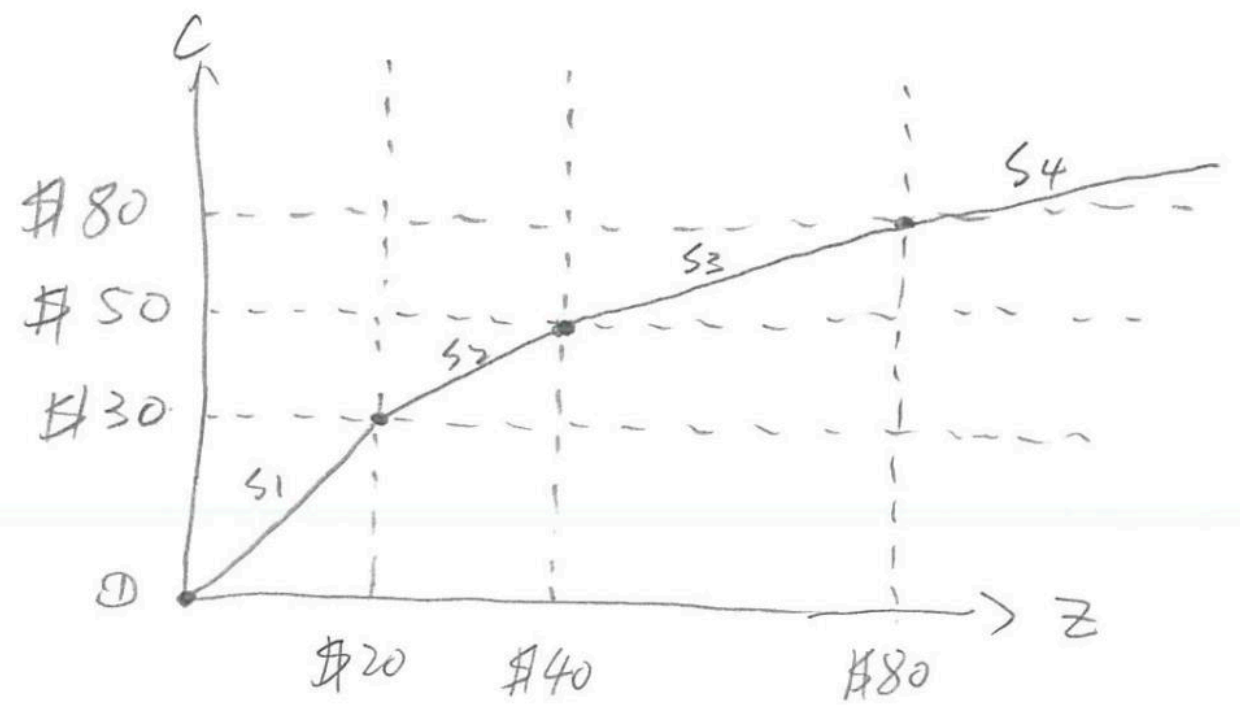
(d) The city replaces the NIT with an **Earned Income Tax Credit (EITC)**:

- **No payment to non-workers**: $c(0) = 0$.
- **Phase-in** ($z = 0$ to \$20): workers receive a subsidy of \$0.50 per dollar earned.
- **Plateau** ($z =$ \$20 to \$40): workers receive a flat credit of \$10.
- **Phase-out** (z above \$40): the credit is reduced by \$0.25 per dollar of earnings above \$40.
- No other taxes.

At what earnings level is the credit fully phased out? Draw the budget constraint (same axes as part (a)). Label the y-intercept, all kink points (with coordinates), and the slope of each segment. *(12 points)*

\$10 ÷ \$0.25 = \$40 , \$40 + \$40 = \$80

when z ≥ \$80, the credit fully phased out



$S_1 > S_2 > S_3 > S_4$

(slope.)

(e) Motivated by the EITC structure in part (d), the government considers a modest reform to the NIT from part (c): it **reduces the phase-out rate from 50% to 25%** on the $\boxed{\text{first \$20}}$ of earnings. Above \$20, the phase-out rate remains 50%. All else stays the same (the guaranteed income is still \$20). *(12 points total)*

(i) What is the participation tax rate at $z = 20$ under this reformed NIT? Compare it to the original NIT participation tax rate from part (b). *(4 points)*

$$\tau_p' = \frac{T(20) - T(0)}{20} = \frac{5}{20} = 0.25$$

$$T_p = 0.5 \rightarrow \tau_p' = 0.25 \text{ now.}$$

the participation tax rate at $z = 20$ decrease half as before.

$\Rightarrow$ more workers stay

(ii) The reform has a **mechanical cost**: the 100 workers at $z = 20$ who were already working now receive a larger net transfer. How much extra does each one cost the government? What is the total mechanical cost? *(4 points)*

previously = \$10 transfer (50% phase out)

now : \$ 15 transfer (25% phase out)

(\$15 − \$10) × 100 = \$500 → total mechanical cost.

\$5 → extra on each worker

(iii) Suppose that among the 200 residents with potential earnings $z = 20$, the **extensive-margin elasticity of the number who work** with respect to the net-of-tax rate $1 - \tau_p$ is

$$\varepsilon = \frac{\%\Delta\text{workers}}{\%\Delta(1-\tau_p)} = 2.$$

Using the original NIT in part (c) as the baseline, compute how many of the 100 workers who had dropped out would re-enter when the participation tax rate falls from 50% to 25%. Then compute the **fiscal saving** from each returning worker: compare what they cost as a non-worker versus what they cost as a worker under the reformed NIT. What is the total extensive-margin saving? Does it cover the mechanical cost from part (ii)? Explain in 1–2 sentences why this type of reform can be a "free lunch." *(4 points)*

extensive → work or not.

$\Rightarrow \tau_p = 50\% = 0.5 \rightarrow \varepsilon = \frac{\%\Delta \text{ workers}}{0.5} = 2. \rightarrow \%\Delta \text{ workers} = 4\%$.

$\Rightarrow 100 \times 4\% = 4$ workers

Reminder: please sign the Academic Honesty Pledge on the first page!

Scratch Work

If you want anything on this page to be graded, clearly indicate in the relevant question where your answer continues.

Scratch Work

If you want anything on this page to be graded, clearly indicate in the relevant question where your answer continues.